Ramadan Mohamed Hassan

AI Engineer | AI Automation | LLM & RAG Systems | Machine Learning

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LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

AI Engineer and recent B.Sc. graduate in Computer Science – Artificial Intelligence, specializing in production-oriented AI systems, automation workflows, machine learning, computer vision, LLM applications, RAG, and backend AI services. Experienced in developing FastAPI services, n8n automations, API integrations, data pipelines, and deployable AI components. Co-author of research published in Elsevier's Q1 Internet of Things journal and Springer proceedings, with award-winning projects across smart agriculture, medical AI, robotics, and optimization.

EDUCATION

New Mansoura University | B.Sc. Computer Science - Artificial Intelligence Program | Sep 2022 - Jun 2026

PROFESSIONAL EXPERIENCE

Rawi Digital Transformation & Services — AI & Machine Learning Engineer (Full-Time, Remote) *Jun 2026 – Present*

- Build AI automation solutions and LLM-based workflows using n8n, FastAPI, REST APIs, RAG, and vector databases to support digital transformation and internal business processes.
- Develop backend AI services, reporting automations, API integrations, and data pipelines that connect business systems with machine learning and LLM capabilities.

Al Osrah Group — Healthcare & Medical Services — AI Engineer (Part-Time, On-site) *Jun 2026 – Jul 2026*

- Developed machine learning models and internal automation tools for pharmacy, medical-distribution, and laboratory operations.
- Applied OCR and LLM-assisted workflows to digitize records and support reporting, inventory analysis, and demand-pattern evaluation.

ReNile Company — Artificial Intelligence Engineer Intern (On-site) *Mar 2026 – May 2026*

- Contributed to deployment-oriented AI projects covering data preprocessing, model optimization, and practical machine learning implementation.
- Developed deployable AI components supporting solutions used by approximately 70% of ReNile's client base, based on internal company estimates.

RESEARCH & PUBLICATIONS

- **Internet of Things – Elsevier | Volume 39 | Article 102025 | 2026 | Q1 Journal – SCImago 2025**
  "An Edge–Cloud AI–Robotics Framework with Blockchain-Based Validation for Crop Yield Prediction and Plant Health Monitoring."
  Co-developed an edge-cloud smart-agriculture framework integrating AI, autonomous robotics, UAV imagery, IoT sensor networks, and blockchain-backed validation for crop-yield prediction and plant-health monitoring. Publication
- **Springer Conference Proceedings – ISBCOM 2025 | 2025**
  "Hybrid Random Walk-Dijkstra Approach for Efficient Coverage Path Planning in Robotics."
  Developed a robotic path-planning approach combining Random Walk, Dijkstra's algorithm, and computer vision for obstacle-aware navigation. Publication

SELECTED TECHNICAL PROJECTS

- **AgriNova – AI, Robotics, IoT & Blockchain Smart Agriculture System**
  Built a smart-agriculture platform integrating VGG16 disease detection (96.9% accuracy), IoT monitoring, UAV/robotics, blockchain traceability, and analytics. The project formed the basis of research published in Elsevier's Q1 Internet of Things journal and won 1st Place at CSE 2026 and 2nd Place at MansTech, with a 30,000 EGP prize. GitHub
- **Brain Tumor MRI AI System**
  Built a CNN-based medical-imaging pipeline for automated brain-tumor classification from MRI scans using TensorFlow/Keras, preprocessing, model training, and evaluation workflows. GitHub
- **Exoplanet Detection ML System**
  Developed a scientific machine-learning pipeline for detecting planetary transits in noisy light-curve data, including model comparison and reproducible evaluation. GitHub

- **Baby Food Nutrition Recommendation System**
  Developed a TabNet-based infant-nutrition recommendation system analyzing 26 nutrition and macronutrient features and reporting 99% accuracy for personalized diet suggestions. GitHub
- **Hybrid Optimization Algorithm**
  Created an award-winning hybrid optimization approach combining metaheuristic search strategies and benchmarking on standard optimization functions; recognized 1st Place among 45 university projects. GitHub

AWARDS & CERTIFICATIONS

AWARDS & RECOGNITION

- **2nd Place — MansTech Hackathon, Apr 2026**
  Won 30,000 EGP for an AI, robotics, and IoT agricultural solution endorsed by ITIDA.
- **1st Place — Computer Science and Engineering Projects Exhibition, 2026**
  Awarded for the AgriNova smart-agriculture system.
- **1st Place — Annual University Projects Exhibition, 2025**
  Recognized for an innovative optimization algorithm among 45 projects.
- **ICPC & ECPC Qualification, 2023**
  Ranked among the top 15% of participating teams.

CERTIFICATIONS & TRAINING

- **NVIDIA** — Building LLM Applications with Prompt Engineering; AI for All: From Basics to GenAI Practice.
- **Huawei ICT Academy** — HCIA AI; HCIP AI.
- **NTI / Huawei Egyptian Talent Academy** — 80-hour Artificial Intelligence Training, 97% score.

TECHNICAL SKILLS

Programming: Python, JavaScript/TypeScript, SQL

AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, NLP, Predictive Modeling, Feature Engineering, Model Evaluation

LLM & AI Systems: LLM Applications, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, LangChain, LangGraph

Frameworks & Libraries: PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, Hugging Face Transformers, Pandas, NumPy

Backend & Automation: FastAPI, REST APIs, n8n, Workflow Automation, API Integrations, Streamlit

Data & Deployment: PostgreSQL, MongoDB, FAISS, Pinecone, Git/GitHub, Vercel

COURSES & LANGUAGES

Courses: Computer Vision Specialization — Coursera; NLP with Deep Learning — Coursera.

Languages: Arabic — Native **|** English — Professional Working Proficiency